



Project proposal

On

4 bit comparator with display

Course No: ECE 2104

Course Name: Digital Electronics and Logic Circuit

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Project title: 4 bit comparator with display .

Circuit diagram:

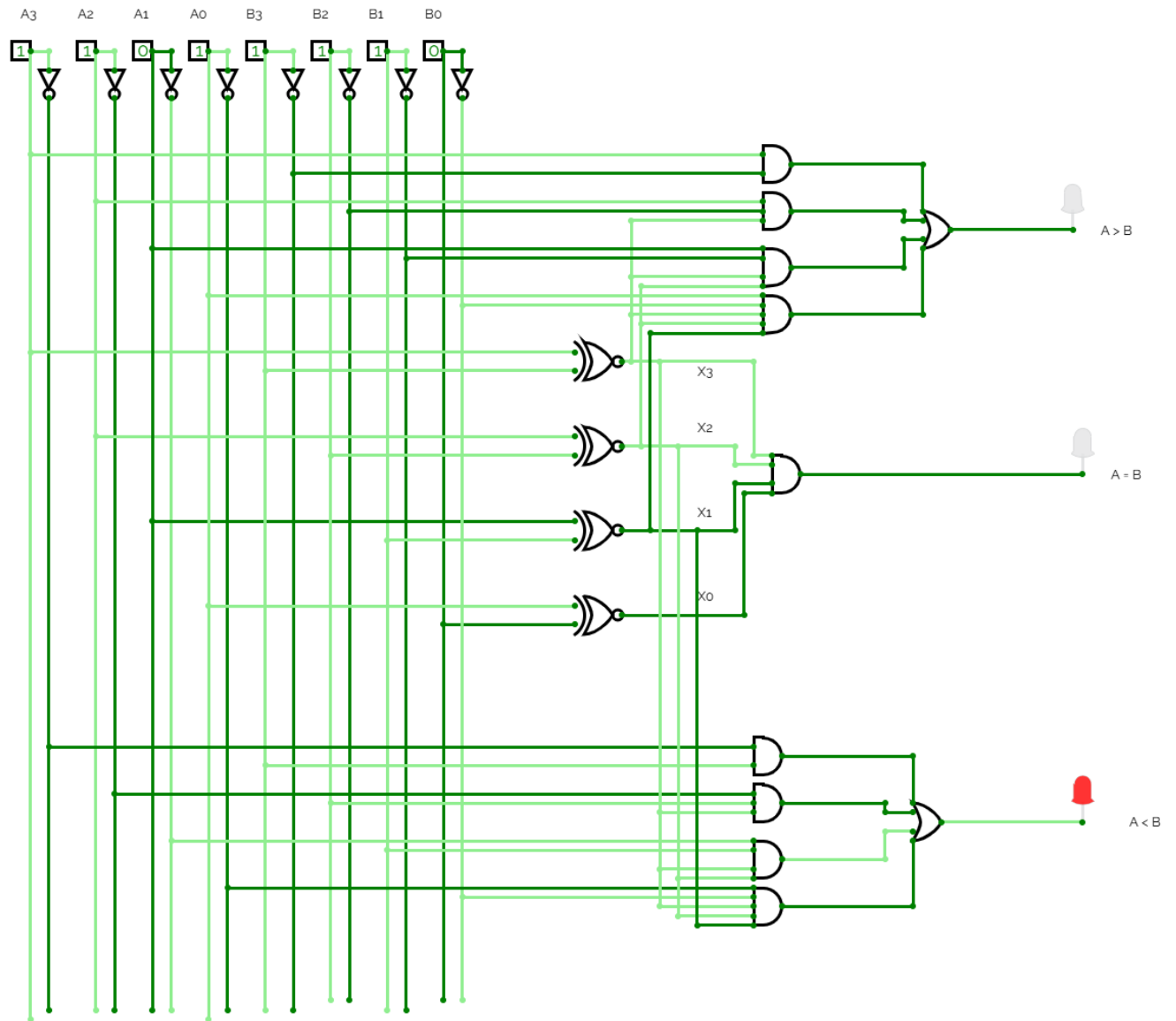


Fig: Circuit Diagram for 4 bit comparator circuit.

Required circuit components:

SL No.	Circuit Component	Ratings / Specification	Quantity
1	IC 7404 (NOT Gate)	Hex Inverter, 5V TTL	2
2	IC 7408 (2-input AND)	Quad AND Gate	2
3	IC 7411 (3-input AND)	Triple AND Gate	2
4	IC 7421 (4-input AND)	Dual AND Gate	3
5	IC 4072 (4-input OR)	Dual OR Gate (CMOS)	2
6	IC 7486 (XOR Gate)	Quad XOR	2
7	7-Segment Display	Common Anode/Cathode	1
8	IC 7447 / 4511	BCD to 7-Segment Driver	1
9	Resistor	220 Ω –330 Ω	8
10	LED	5mm	3
11	Resistor (LED)	220 Ω	3
12	Toggle Switch	5V compatible	8
13	Breadboard	Standard	1
14	Connecting Wires	Jumper	As required
15	DC Power Supply	+5V	1

Expected outcome :

The proposed project aims to design and implement a functional 4-bit digital comparator circuit using logic gates. The expected outcome is a fully operational system capable of comparing two 4-bit binary inputs (A and B) and accurately determining their relationship.

The circuit will generate three distinct outputs indicating whether $A > B$, $A = B$, or $A < B$. These outputs will be visually represented using LEDs and further displayed through a 7-segment display for clear and user-friendly interpretation.

By the completion of the project, the following outcomes are expected:

- Successful construction of the comparator circuit using basic logic gate ICs.
- Accurate comparison of two 4-bit binary numbers in real-time.
- Proper functioning of output indicators ($A > B$, $A = B$, $A < B$).
- Integration of a 7-segment display to present the result in a readable format.
- Improved understanding of digital logic design, gate-level implementation, and practical circuit building.

Overall, the project will demonstrate the practical application of combinational logic design and provide hands-on experience in implementing digital systems using hardware components.

Details of the approach :

The proposed project will be implemented through a structured hardware design methodology using basic digital logic components. The approach focuses on designing a 4-bit comparator at the gate level and verifying its operation through practical implementation.

Initially, the logical relationship between two 4-bit binary numbers ($A = A_3, A_2, A_1, A_0$ and $B = B_3, B_2, B_1, B_0$) will be analyzed. Boolean expressions for the three output conditions ($A > B$, $A = B$, and $A < B$) will be derived using fundamental principles of combinational logic. Equality between corresponding bits will be determined using XOR/XNOR operations, while magnitude comparison will be established using AND, OR, and NOT gate combinations.

In the next step, the derived Boolean expressions will be converted into a complete gate-level circuit diagram. Multi-input logic requirements (such as 3-input, 4-input, and 5-input AND/OR operations) will be implemented using ICs or by combining lower-input gates where necessary.

Following the design phase, the circuit will be physically constructed on a breadboard. Required ICs (such as 7408, 7411, 7421, 7486, 7404, and 4072) will be interconnected using jumper wires. Toggle switches will be used to provide input combinations, and a regulated +5V power supply will be used to operate the circuit.

For output visualization, LEDs will be connected to represent the three comparison conditions. Additionally, a 7-segment display will be integrated using an appropriate decoder/driver IC (such as 7447 or 4511) to present the output in a more user-friendly format.

Finally, the system will be tested with all possible input combinations to verify correctness and reliability. Any discrepancies will be debugged, and necessary adjustments will be made to ensure accurate performance.